

HW10

P1. (a)(i) $x(j\omega) = \frac{1}{(2+j\omega)^2}$ $H(j\omega) = \frac{1}{j\omega+4}$

$$Y(j\omega) = x(j\omega) \cdot H(j\omega) = \frac{1}{(2+j\omega)^2(j\omega+4)} = \frac{-\frac{1}{4}}{j\omega+2} + \frac{\frac{1}{2}}{(j\omega+2)^2} + \frac{\frac{1}{4}}{j\omega+4}$$

$$Y(t) = \left[-\frac{1}{4}e^{-2t} + \frac{1}{2}te^{-2t} + \frac{1}{4}e^{-4t} \right] u(t)$$

(ii) $x(j\omega) = \frac{1}{(j\omega+2)^2}$ $H(j\omega) = \frac{1}{(j\omega+4)^2}$

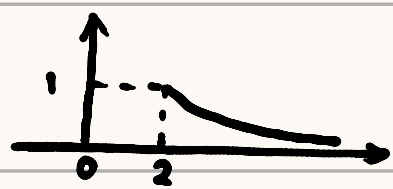
$$Y(j\omega) = \frac{1}{(j\omega+2)^2(j\omega+4)^2} = \frac{-\frac{1}{4}}{j\omega+2} + \frac{\frac{1}{4}}{(j\omega+2)^2} + \frac{\frac{1}{4}}{j\omega+4} + \frac{\frac{1}{4}}{(j\omega+4)^2}$$

$$Y(t) = \left[-\frac{1}{4}e^{-2t} + \frac{1}{4}te^{-2t} + \frac{1}{4}e^{-4t} + \frac{1}{4}te^{-4t} \right] u(t)$$

(iii) $x(j\omega) = \frac{1}{j\omega+1}$ $H(j\omega) = \frac{1}{-j\omega+1}$

$$Y(j\omega) = \frac{1}{1+\omega^2}$$

$$Y(t) = \frac{1}{2}e^{-|t|}$$

(b) $x(t) = e^{-(t-2)}u(t-2)$ 

$$x(t) * h(t) = \int_2^{\infty} e^{-(t-\tau)} h(t-\tau) d\tau$$

① $1+t \leq 2 \Rightarrow t \leq 1$ $y(t) = 0$

② $1+t > 2 \Rightarrow 1 < t \leq 5$ $y(t) = e^2 \int_2^{1+t} e^{-\tau} d\tau = e^2 (-e^{-\tau}) \Big|_2^{1+t}$

$$= e^2 [-e^{-1-t} + e^{-2}] = -e^{1-t} + 1$$

$$Y_1(j\omega) = \int_1^5 (-e^{1-t} + 1) e^{-j\omega t} dt = \frac{e^{-j\omega} - e^{-5j\omega}}{j\omega} - \frac{e^{-j\omega} - e^{-4j\omega}}{1+j\omega}$$

③ $5 < t$ $y(t) = e^2 \int_{t-3}^{1+t} e^{-\tau} d\tau = e^2 [-e^{-\tau} + e^{-t+3}] = -e^{1-t} + e^{-t+5}$

$$Y_2(j\omega) = \int_5^{\infty} -e^{1-t} + e^{-t+5} dt = \frac{e^{-5j\omega} - e^{-4j\omega}}{1+j\omega}$$

$$Y_1(j\omega) + Y_2(j\omega) = \frac{e^{-j\omega} - e^{-5j\omega}}{j\omega(1+j\omega)}$$

$$x(j\omega) = \frac{e^{-2j\omega}}{1+j\omega} \quad H(j\omega) = \int_{-1}^3 e^{-j\omega t} dt = \frac{e^{j\omega} - e^{3j\omega}}{j\omega}$$

$$Y(j\omega) = \frac{e^{-j\omega} - e^{-5j\omega}}{j\omega(1+j\omega)}$$

$\therefore$ two ways are the same

P2. (a) $[(j\omega)^2 + j\omega + 1] Y(j\omega) = (j\omega + 2) X(j\omega)$

$$H(j\omega) = \frac{j\omega + 2}{(j\omega)^2 + j\omega + 1} = \frac{1}{1+j\omega} + \frac{1}{(j\omega+1)^2}$$

$$h(t) = [e^{-t} + te^{-t}] u(t)$$

(b) $[(j\omega)^3 + 4(j\omega)^2 + 5j\omega + 2] Y(j\omega) = [(j\omega)^2 + 3] X(j\omega)$

$$H(j\omega) = \frac{(j\omega)^2 + 3}{(j\omega)^3 + 4(j\omega)^2 + 5j\omega + 2} = -\frac{6}{j\omega+1} + \frac{4}{(j\omega+1)^2} + \frac{7}{j\omega+2}$$

$$h(t) = [-6e^{-t} + 4te^{-t} + 7e^{-2t}] u(t)$$